

Run 12 | NME

India

IV + controls

Target Necessity-motivated entrepreneurship; binary 1 vs. 0
Sample 27,555 respondents **Coverage** 1 country; 2015–2022
Positive cases 1,849 (6.71%) **Case time** 1m 22s
Train / val. / test 18,740 / 3,305 / 5,510 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.
Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 26.
Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: H2O GBM (0.9879). Highest F1: CatBoost (0.7664). For H2O GBM, adding controls changed ROC-AUC by -0.0002 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
H2O GBM	0.9879	0.8301	96.46	69.59	83.74	0.7601	4s
CatBoost	0.9878	0.8238	96.82	75.53	77.78	0.7664	21s
XGBoost	0.9875	0.8094	96.82	75.66	77.51	0.7657	<1s
Lasso	0.9866	0.8050	96.62	73.52	77.51	0.7546	15s
Random Forest	0.9866	0.8179	96.21	67.32	84.28	0.7485	1s
Elastic Net	0.9865	0.8043	96.59	73.39	76.96	0.7513	15s
Logistic Regression	0.9863	0.8027	96.66	73.90	77.51	0.7566	2s
Ridge	0.9859	0.8078	96.55	72.77	77.51	0.7507	12s
Decision Tree	0.9857	0.7648	96.68	76.27	73.17	0.7469	<1s
Naive Bayes	0.8684	0.4606	93.19	49.29	56.64	0.5271	7s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	617.57	41.32
2	Established business owner	233.47	15.62
3	Occupational status	73.18	4.90
4	Age	59.11	3.96
5	Household size	41.26	2.76
6	Knowing entrepreneurs	31.71	2.12
7	Household income	29.19	1.95
8	Education attainment	24.72	1.65

Share is the percentage of total forest impurity importance across all 66 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Market access and infrastructure	9.61	0.64
2	Governance	9.45	0.63
3	Safety and security	9.16	0.61

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Source: Completed Objective 2 batch, 28 September 2026; 12_obj2_india_iv_cv_NME. Scores and importance are taken directly from saved CSV outputs.